

Executive Summary

Dedicated professional at the intersection of Urban Planning, Spatial Data Science, Transport Studies, and Civil Engineering, currently pursuing a Master's in Urban Spatial Science at UCL. Expert in using advanced spatial analysis and machine learning for designing resilient and sustainable cities. Proven track record in leading significant projects, including the £15 million National Road Program in India. Academic projects span strategic route planning, crop yield estimation, urban heat island studies, and impact assessments using satellite data. Committed to transforming urban environments by enhancing transport connectivity, shaping public policy, and improving city resilience in line with the UN Sustainable Development Goals.

Educational Information

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| University College London (UCL) | London, UK |
| <ul style="list-style-type: none">• MSc, Urban Spatial Science.• Commonwealth Scholar -23/24• Bartlett Student Ambassador 23/24 | (2023-2024) |
| National Institute of Technology (NIT) | Srinagar, India |
| <ul style="list-style-type: none">• Bachelor of Technology (BTech), Civil Engineering.• Achieved highest SGPA of 9.68 among over 1000 Engineering students in the first year.• Silver Medalist with a CGPA of 9.241/10 among 120 students. | (2009-2013) |

Experience

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| Public Works Department (JKPWD), Govt. of India | India |
| Assistant Engineer, PMGSY | Mar 2017 – Aug 2023 |
| <ul style="list-style-type: none">• Led the implementation of the Indian National Rural Road Plan (PMGSY-III) to connect rural populations in Northern Region of India, managing spatial planning, groundwork, & implementation.• Ensured effective communication within the team & served as a liaison between multiple stakeholders.• Recognized as "Outstanding Officer" by the Government of J&K, India for 2021-22. | |
| Indian Oil (IOCL), Ministry of Petroleum and Natural Gas, Govt. of India | India |
| Officer (Health, Safety, Engineering and Environment) | May 2015 – Feb 2017 |
| <ul style="list-style-type: none">• Conducted environmental and safety audits, managed work permit issuance, and led job safety analysis (JSA) for engineering and operational activities at major oil installations.• Framed the Disaster Management Plan for the Oil Installations.• Recognized with "Outstanding Employee" award for 2015-16 [Performance rating of 4.38 out of 4.5]. | |

Projects

- **Urban Heat Island Dynamics in London Wards: The Role of Built Environment Features:** Investigated the impact of built environment characteristics on the Urban Heat Index using data from the London Datastore and Google Earth Engine. Employed multiple linear regression to analyze how various features influence UHI across different London wards.
- **Strategic Route Planning and Economic Analysis for Delivery Services of a City:** Developed a NetLogo model to optimize operational strategies and financial performance using real-world geographic data.
- **Crop Yield Estimator:** Conducted a detailed analysis of crop yield using historical data and machine learning algorithms. Employed techniques such as linear regression, decision trees, and random forests to predict crop yield and identify key influencing factors.
- **Investigating Survival Disparity:** Conducted a comparative analysis of cities with unequal resources using the Sugarscape models.
- **Analyzing Spatial Clustering of Public Service Facilities in Andhra Pradesh, India:** Explored the spatial clustering of public service facilities post-bifurcation using K-means, Hierarchical, and DBSCAN clustering methods. Analyzed the distribution of health, transport, education, and agricultural services at the block level, identifying regional disparities and service density variations.
- **Machine Learning for Temporal Change Detection in LULC:** Applied CART and Random Forests using Google Earth Engine for landcover classification in Srinagar City.
- **Comparing Socio-spatial Traits in London Neighbourhoods:** Analyzed the socio-spatial characteristics of Whitechapel and Canary Wharf using Soja's socio-spatial dialectic.

Achievements

- Selected for three prestigious Russell Group Universities in the UK with full scholarships: Tetley and Lupton Scholarship (2023), International Masters Excellence Scholarship (2023), FELS - Southampton Environmental & Life Sciences Dean's International Scholarship, and Commonwealth Scholarship (2023-24).
- Awarded Science and Technology Merit Scholarship by the Dept. of Science & Technology, J&K (2006-2008).
- Received Certificate of Merit by British Safety Council (2017) for commitment to high standards of health, safety, and environment.
- Received Certificate of Appreciation by Govt. of J&K for leadership and effective communication with the Fire Department during an off-site fire and emergency incident.

Other Information

- **Skills:**
 - Q-GIS, Net Logo, Python, R, SQL, Quarto, Git, and Google Earth Engine.
- **Languages:**
 - English (Fluent), Kashmiri (Fluent), Hindi (Fluent), and Urdu (Fluent).
- **Memberships:**
 - Engineering for Change (E4C); Institution of Engineers of India.
- **Volunteering:**
 - Indian Red Cross Society, Volunteer Centre Camden, London, UK.